



**UNIVERSIDAD NACIONAL AUTÓNOMA DE
MÉXICO**

FACULTAD DE ESTUDIOS SUPERIORES ARAGÓN

INGENIERÍA EN COMPUTACIÓN

Estructura de datos

Tarea 12

Prof. Jesús Hernandez Cabrera

Grupo: 1360

Arroyo Gutierrez Rodrigo Javier

```
ject v
mergeSort C:\Users\rodril\IdeaProjects\mergeSort
  .idea
  > out
  > src
    MergeSort
    .gitignore
    mergeSort.iml
  External Libraries
  Scratches and Consoles

MergeSort.java x
3 public class MergeSort {
4
5 @ public static int[] mergeSort(int[] data) { 3 usages
6     if (data.length > 1) {
7         int mitad = data.length / 2;
8         int[] izquierda = Arrays.copyOfRange(data, from: 0, mitad);
9         int[] derecha = Arrays.copyOfRange(data, mitad, data.length);
10        System.out.println(Arrays.toString(izquierda) + " --- " + Arrays.toString(derecha));
11        mergeSort(izquierda);
12        mergeSort(derecha);
13    }
14 }

MergeSort x
:
:
:
...-.-.-.- MERGE -.-.-.-
[38, 27, 43, 3, 9] --- [82, 10, 19, 50, 61]
[38, 27] --- [43, 3, 9]
[38] --- [27]
regreso de rec: [38]
regreso de rec: [27]
regreso de rec: [27, 38]
[43] --- [3, 9]
regreso de rec: [43]
[3] --- [9]
regreso de rec: [3]
regreso de rec: [9]
regreso de rec: [3, 9]
regreso de rec: [3, 9, 43]
regreso de rec: [3, 9, 27, 38, 43]
[82, 10] --- [19, 50, 61]
[82] --- [10]
```

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MergeSort x
:
:
:
[82, 10] --- [19, 50, 61]
[82] --- [10]
regreso de rec: [82]
regreso de rec: [10]
regreso de rec: [10, 82]
[19] --- [50, 61]
regreso de rec: [19]
[50] --- [61]
regreso de rec: [50]
regreso de rec: [61]
regreso de rec: [50, 61]
regreso de rec: [19, 50, 61]
regreso de rec: [10, 19, 50, 61, 82]
regreso de rec: [3, 9, 10, 19, 27, 38, 43, 50, 61, 82]
[3, 9, 10, 19, 27, 38, 43, 50, 61, 82]

Process finished with exit code 0
```